



CHRIST

(DEEMED TO BE UNIVERSITY)

PUNE LAVASA CAMPUS

The Hub of Analytics

School of Commerce, Finance & Accountancy

M.Sc. Global Finance & Analytics Program

Data Analysis and Machine Learning

Mini Project Data Analysis

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1. Introduction

The global surge in artificial intelligence has triggered a massive capital rotation into semiconductor technologies, positioning NVIDIA Corporation (NVDA) at the epicenter of a historic market super-cycle. Analyzing this asset provides a unique opportunity to study extreme market momentum, volatility clustering, and algorithmic price behavior.

Problem Statement: The highly volatile and sentiment-driven nature of AI stocks makes it difficult to consistently forecast daily directional price movements and future valuations using traditional financial models.

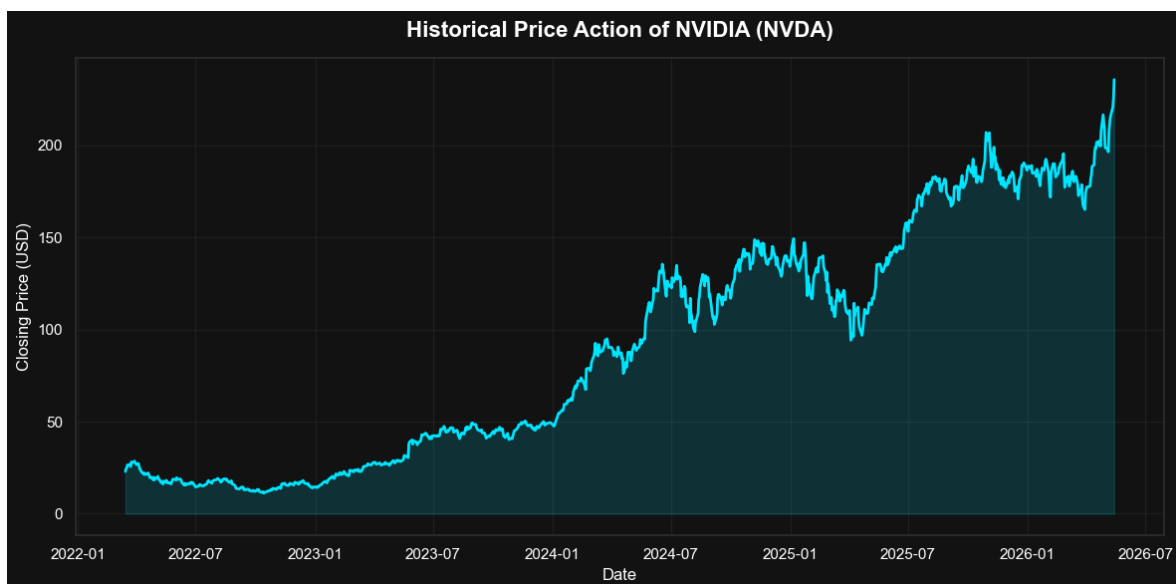
Objective: To utilize machine learning to classify daily directional market movements and accurately predict the next-day closing price of NVIDIA stock.

2. Description of the Data and Origin

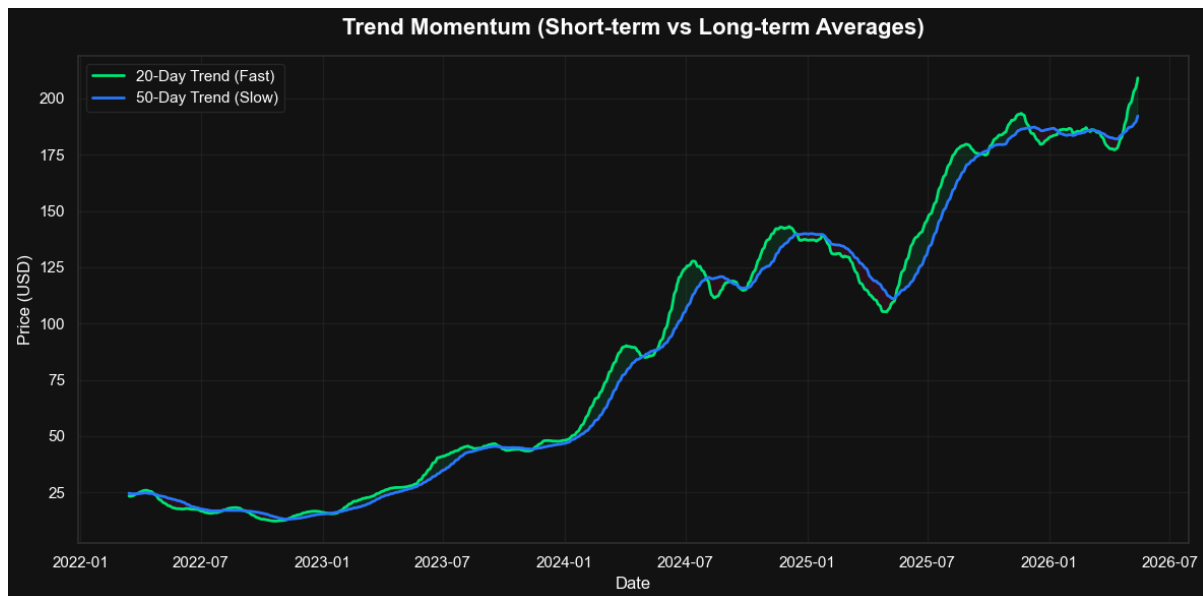
The study utilizes real-time, daily financial time-series data extracted programmatically via the yfinance API. The primary asset is NVIDIA (NVDA), with the NASDAQ 100 Index (^NDX) used for benchmark integrations.

- i. **Timeframe:** March 15, 2022, to May 14, 2026.
- ii. **Volume:** 1,074 total trading days analyzed (1,046 days used for regression models after preprocessing).
- iii. **Variables:** Open, High, Low, Close, and Trading Volume.

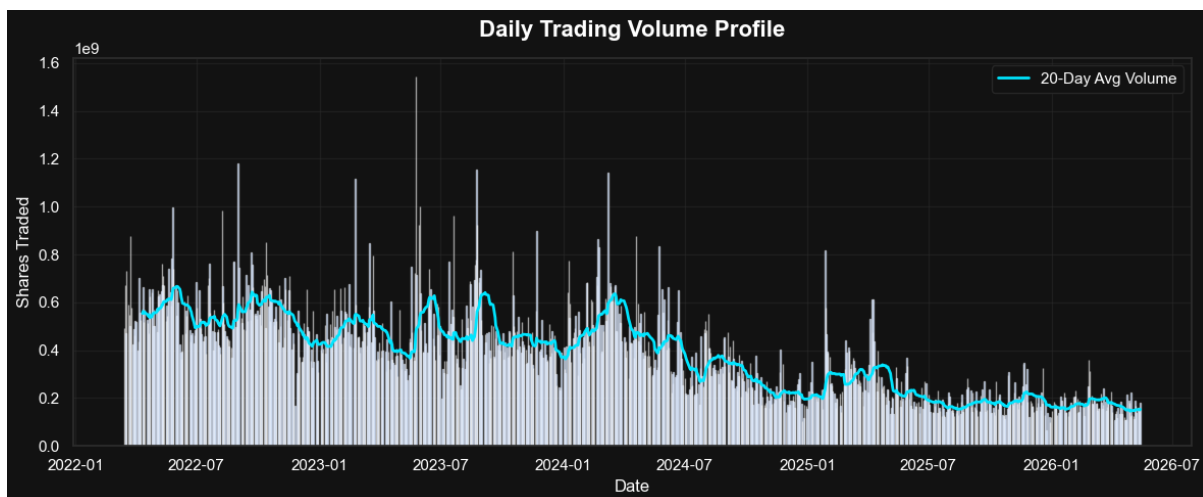
3. Exploratory Data Analysis with Visualizations and Inferences



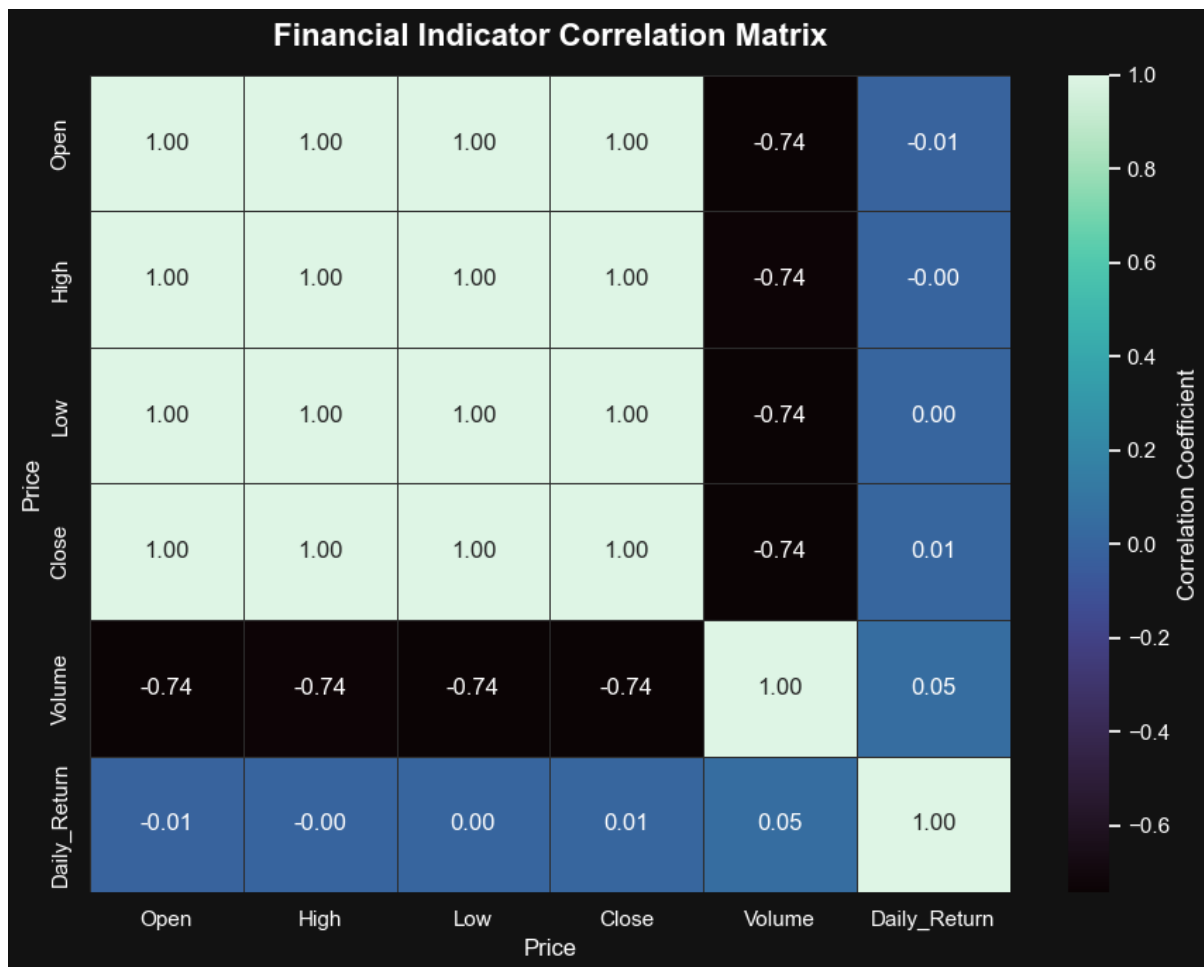
Inference: The area chart illustrates a parabolic, exponential growth curve reflecting NVIDIA's dominance in the AI hardware sector. The baseline support levels continue to rise, indicating aggressive institutional accumulation.



Inference: The dual-line chart maps the 20-day (fast) and 50-day (slow) Simple Moving Averages. Prolonged periods where the fast trend stays above the slow trend confirm strong, sustained bullish momentum.



Inference: The bar chart overlayed with average volume reveals that massive price gaps are almost entirely supported by significant volume spikes, indicating high market liquidity rather than low-volume manipulation.



Inference: The heatmap confirms strong collinearity between core price indicators (Open, High, Low, Close). However, daily returns exhibit minimal linear correlation with outright volume, suggesting volatility is not exclusively volume-dependent.

4. Preparation of Data through Data Cleaning and Required Manipulations

To prepare the dataset for machine learning, robust preprocessing was executed:

Handling Missing Data:

NaN values generated by API lags and rolling window calculations were systematically dropped to prevent mathematical errors during model training.

Feature Engineering:

- i. Calculated 20-Day and 50-Day Simple Moving Averages (SMA).
- ii. Extracted Daily Percentage Returns.
- iii. Calculated a 20-Day Annualized Volatility metric.

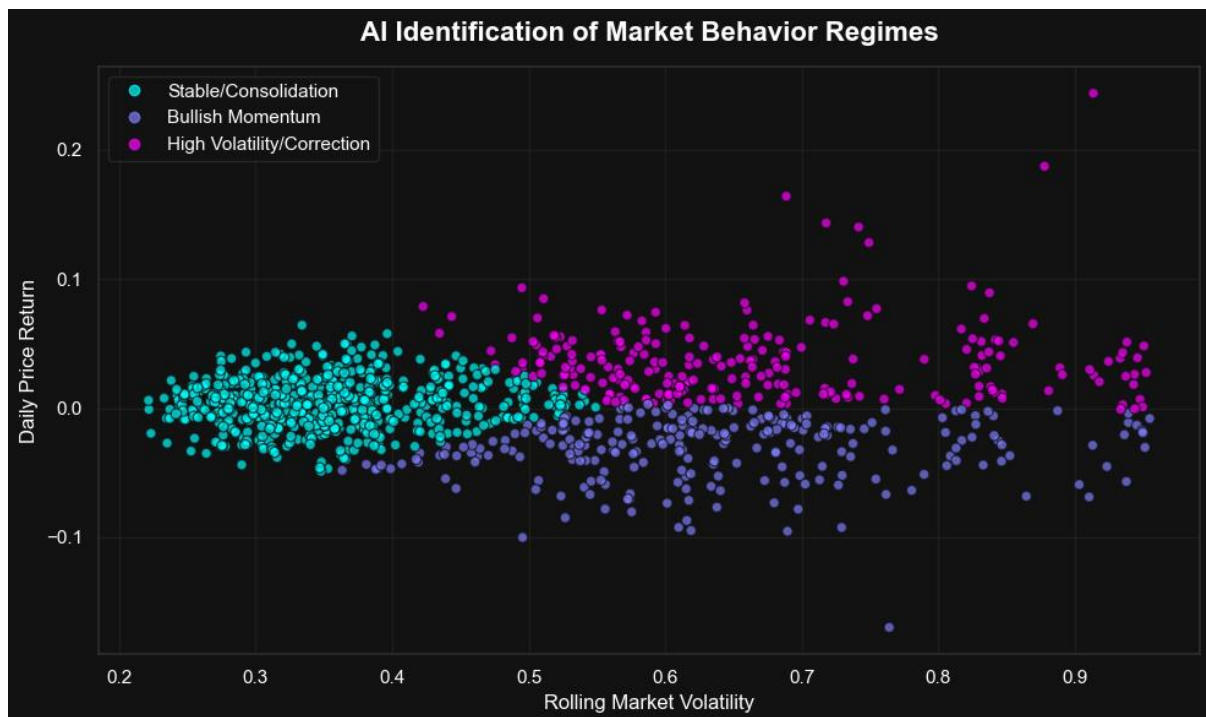
Target Variable Creation:

- i. **For Regression:** Shifted the closing price backward by one day to create a "Next-Day Closing Price" target.
- ii. **For Classification:** Created a binary target where 1 signifies a bullish next day (price goes up) and 0 signifies a bearish next day (price goes down).

5. Predictive Analysis

Part A: Unsupervised Learning of Market Regimes

To understand market conditions, K-Means Clustering was deployed on Volatility and Daily Returns.

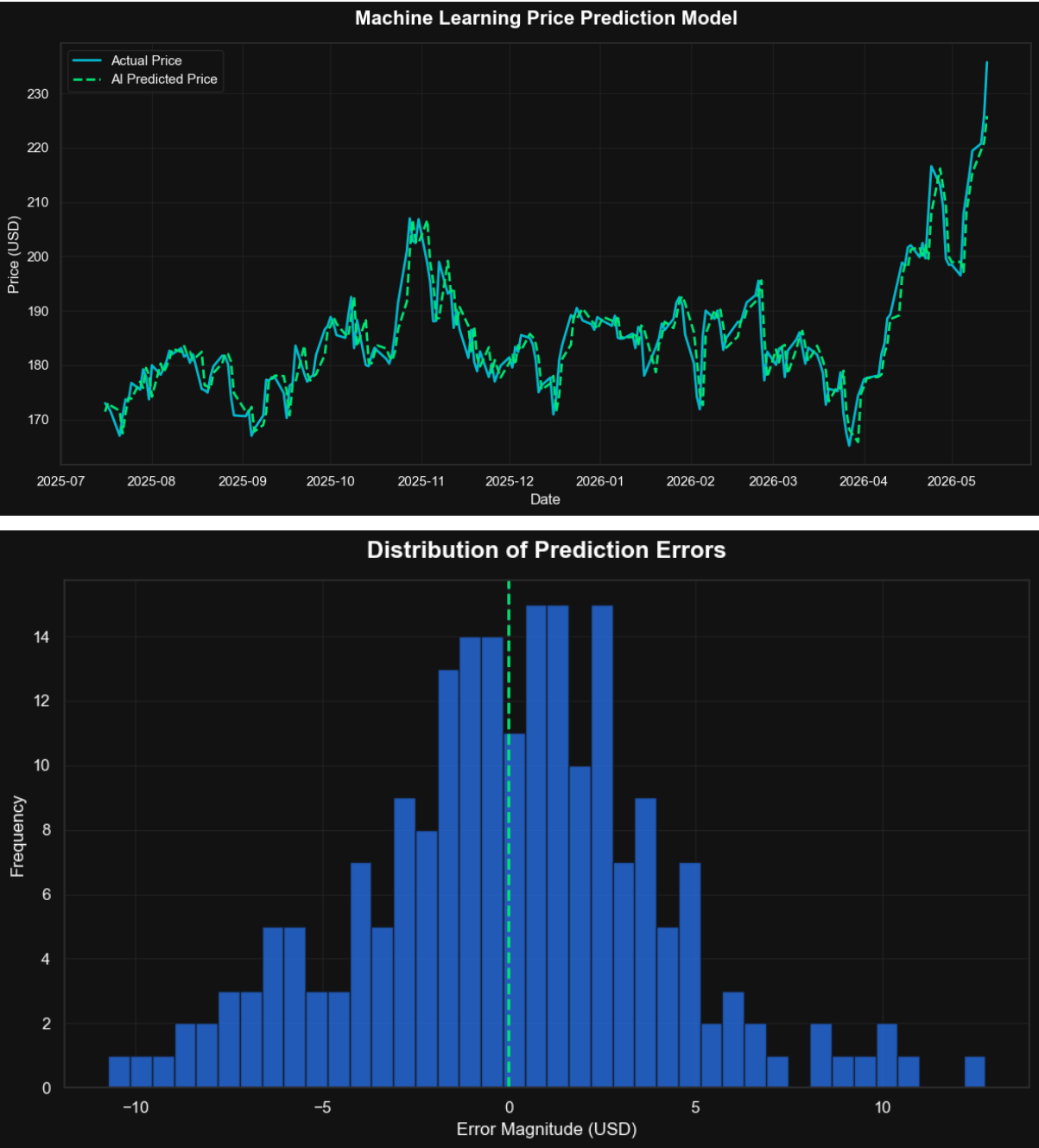


Analysis: The algorithm successfully segmented the 1,046 trading days into three distinct behavioral regimes:

- i. Regime 0 (563 days of standard consolidation/growth)
- ii. Regime 1 (251 days of heightened momentum)
- iii. Regime 2 (232 days of extreme volatility).

Part B: Supervised Learning through Regression - Price Prediction

A Multiple Linear Regression model was deployed to predict the exact next-day closing price.

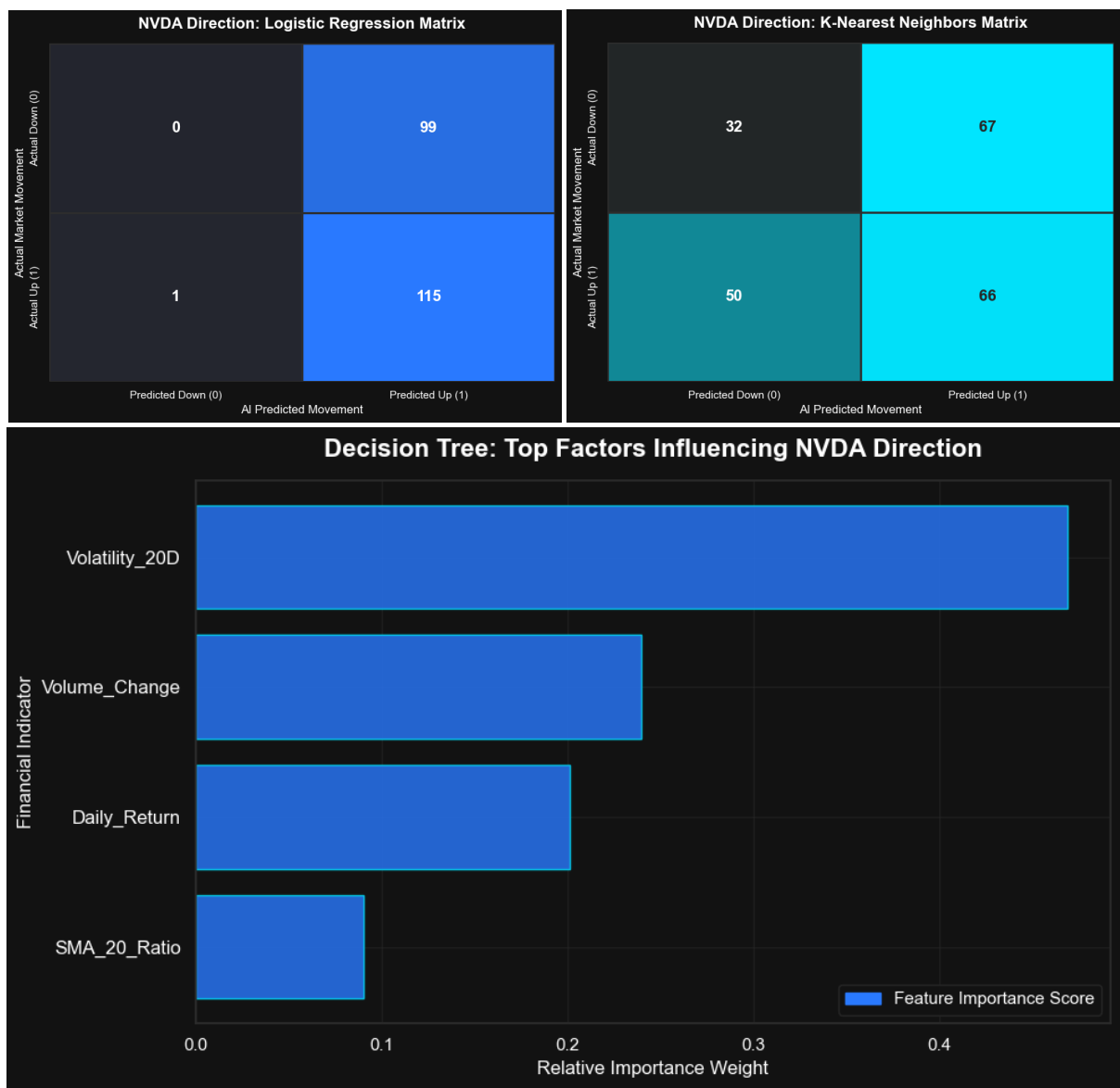


Accuracy & Scores: The model achieved an outstanding **R-Squared (R2) accuracy of 0.8661**, explaining 86.6% of the variance in future prices. The Mean Squared Error (MSE) settled at 16.35.

Notable Factors: Feature weight analysis revealed that the current Close price strongly dictates the next (Weight: 0.9664), while Volatility_20D acts as a negative penalizing factor (Weight: -0.1570).

Part C: Supervised Learning through Classification - Directional Prediction

Three models were tested to predict if tomorrow's price would go UP or DOWN.



Accuracy & Scores: Tested on 215 chronological future days, **Logistic Regression** performed the best with **53.49% accuracy**, followed by Decision Trees (52.56%) and KNN (45.58%).

Analysis: As seen in the confusion matrices, predicting daily direction is notoriously difficult. The Logistic Regression report reveals a "Bull Market Bias." Because NVIDIA has been aggressively rising, the model correctly identified 99% of "Up" days (Recall: 0.99) but struggled heavily to predict down days (Precision: 0.00).

6. Limitations of the Study

- i. **Market Imbalance:** The historic bull run of NVDA creates a highly imbalanced dataset. Classification models become biased toward predicting "Up" days simply because they happen more frequently.
- ii. **External Shocks:** The models rely entirely on technical data (historical prices) and cannot account for fundamental catalysts like sudden earnings reports, CEO announcements, or macroeconomic interest rate hikes.

7. Recommendations and Suggestions

- i. **For Algorithmic Traders:** Rely on the Regression Model for setting target exit prices (given its 86.6% accuracy), but avoid relying solely on the Classification models for daily "buy/sell" signals due to their slight edge (53.49%) over random chance.
- ii. **For Data Scientists (Company/Domain):** Future iterations of this model must incorporate Sentiment Analysis via Natural Language Processing (NLP). Scraping financial news headlines to gauge retail and institutional sentiment will likely fix the classification models' inability to predict sudden market downturns.

8. Conclusion

This study successfully navigated the complexities of financial data analysis within the context of the AI super-cycle. Through robust exploratory data analysis, the physical characteristics of NVIDIA's historic momentum were mapped. By applying advanced machine learning techniques, we proved that while pinpointing the exact daily directional shift (up/down) is hindered by the random walk nature of markets, predicting near-term price value is highly viable, yielding a predictive accuracy of nearly 87%. Ultimately, this project highlights both the power and the boundaries of technical machine learning models in modern finance.